Regression Analysis

You have a large enough capital to build a factory:

- What kind of product?
- Is there enough demand?
- Market research?
  - Can we explain the variability of demand with some other factors?
  - Can we predict the future demand to justify the construction of a new plant?

1

Translation

- **regression analysis**: bağlanım analizi. *İstatistikte belirli rasgele değişkenlerin başka değişkenlere olan bağıntısının irdelenmesi ve bağımlı değişkenlerin bağımsızlardan kestirimi analizi.* (Fra. analyse de régression)
- **regression curve**: bağlanım eğrisi. *Bir rasgele değişkenin beklentisini bir ya da daha çok başka değişkene doğrusal-olmayan biçimde ilişkilendiren denklem. Bağlanım eşitliğinin grafik gösterimi üç boyutlu olduğunda bağlanım yüzeyi adını alır.* (Fra. courbe de régression)
- **regression equation**: bağlanım denklemi. *Bir rasgele değişkenin beklentisini bir ya da daha çok başka değişkene ilişkilendiren denklem.* (Fra. relation régressive)
- **regression line**: bağlanım doğrusu. *Yalnızca doğrusal bağlanımı gösteren bağlanım eğrisi.* (Fra. ligne de régression)
- **regression**: bağlanım. *Bir değişkeni öteki değişkenlere bağımlı kılma işi ve biçimi.* (Fra. régression)

2

Simple linear regression (1)

- The sample: $\{(x_i, y_i)\}_{i=1}^{n}$

independent variable

dependent variable

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

random deviation or random error

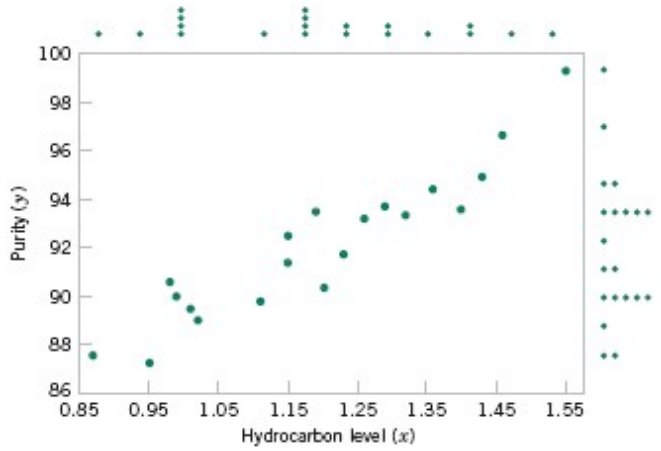
3

Empirical Models

Observation Number	Hydrocarbon Level x(%)	Purity y(%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33

4

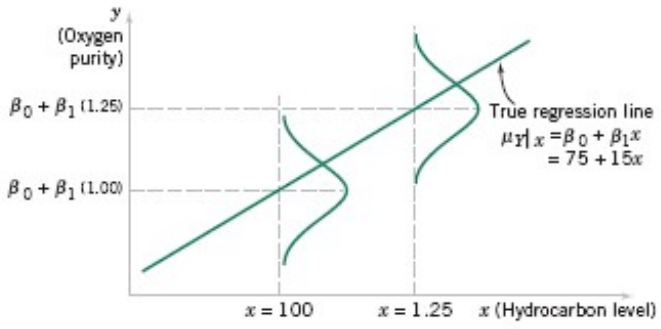
Empirical Models



Scatter Diagram of oxygen purity versus hydrocarbon level.

5

Empirical Models



The distribution of *Y* for a given value of *x* for the oxygen purity-hydrocarbon data.

6

Simple linear regression (2)

$\mu_{Y.x^*}$ expected (or mean) value of Y when x=x*

Expected change in Y with a 1-unit increase in x.

$$\mu_{Y.x^*} = E(\beta_0 + \beta_1 x^* + \varepsilon) = \beta_0 + \beta_1 x^*$$

The mean value of Y is a linear function of x.

$\sigma^2_{Y.x^*}$ variance of Y when x=x*

$$\sigma^2_{Y.x^*} = V(\beta_0 + \beta_1 x^* + \varepsilon) = \sigma^2$$

$$Y \sim N(\beta_0 + \beta_1 x^*, \sigma) \quad \text{at} \quad x^*.$$

7

Estimating model parameters: β_0 & β_1 (1)

$b_0 = \hat{\beta}_0$

$b_1 = \hat{\beta}_1$

$$\hat{\mu}_{Y|X^*=x^*} = b_0 + b_1 x^*$$

$$y_i = \overbrace{b_0 + b_1 x_i}^{\hat{y}_i} + \underbrace{e_i}_{residual}$$

$\hat{y}_i$: estimate for the **i**th observation (x_i,y_i)- fitted or predicted values for x_i

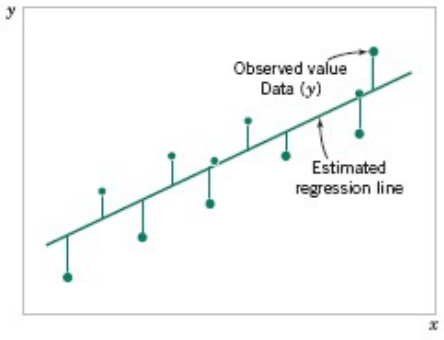
want to minimize the "errors"…

8

Simple Linear Regression

- The method of least squares is used to estimate the parameters, β_0 and β_1 by minimizing the sum of the squares of the vertical deviations in the figure.

Deviations of the data from the estimated regression model.



9

Simple Linear Regression

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

The least squares estimators of β_0 and β_1, say, $\hat{\beta}_0$ and $\hat{\beta}_1$, must satisfy

$$\left.\frac{\partial L}{\partial \beta_0}\right|_{\hat{\beta}_0,\hat{\beta}_1} = -2\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\left.\frac{\partial L}{\partial \beta_1}\right|_{\hat{\beta}_0,\hat{\beta}_1} = -2\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$$

10

Definition: Least square estimates

The **least squares estimates** of the intercept and slope in the simple linear regression model are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}$$

where $\bar{y} = (1/n)\sum_{i=1}^{n} y_i$ and $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$.

11

Fitted or estimated regression line

The **fitted** or **estimated regression line** is therefore

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Note that each pair of observations satisfies the relationship

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + e_i, \qquad i = 1, 2, \ldots, n$$

where $e_i = y_i - \hat{y}_i$ is called the **residual.** The residual describes the error in the fit of the model to the ith observation y_i. Later in this chapter we will use the residuals to provide information about the adequacy of the fitted model.

12

Notation for Simple Linear Regression

$$b_1 = \frac{n\sum x_i y_i - (\sum x_i)(\sum y_i)}{n\sum x_i^2 - (\sum x_i)^2} = \frac{S_{xy}}{S_{xx}}$$

$$b_0 = \overline{y} - b_1\overline{x}$$

$$S_{xy} = \sum_{i=1}^{n}(y_i - \overline{y})(x_i - \overline{x}) = \sum_{i=1}^{n} y_i(x_i - \overline{x})$$
$$= \sum x_i y_i - \frac{\sum x_i \sum y_i}{n}$$

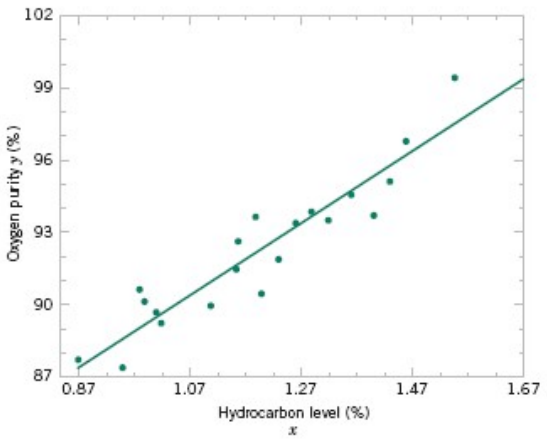
$$S_{xx} = \sum_{i=1}^{n}(x_i - \overline{x})^2 = \sum_{i=1}^{n} x_i(x_i - \overline{x}) = \sum {x_i}^2 - \frac{(\sum x_i)^2}{n}$$

13

Example: Oxygen Purity

What does this equation mean?

Scatter plot of oxygen purity y versus hydrocarbon level x and regression model $\hat{y}$ = 74.20 + 15 x.

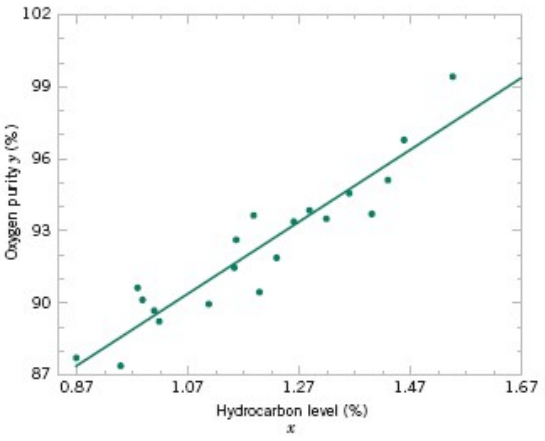


14

Example: Oxygen Purity

If hydrocarbon level (%) is 1, ?

Scatter plot of oxygen purity y versus hydrocarbon level x and regression model $\hat{y} = 74.20 + 15\,x$.



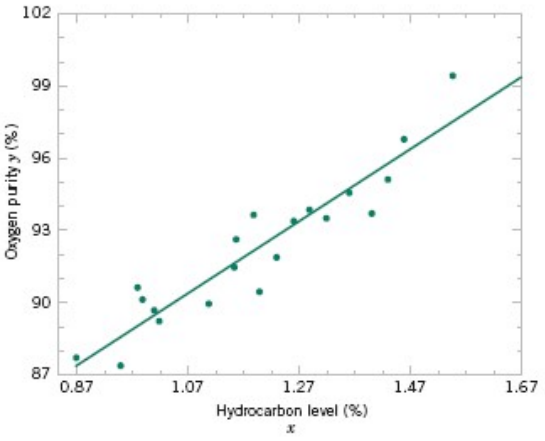
15

Example: Oxygen Purity

extrapolation: dangerous!!!

If hydrocarbon level (%) is 0.2 or 3, ?

Scatter plot of oxygen purity y versus hydrocarbon level x and regression model $\hat{y} = 74.20 + 15\,x$.



16

Example: Oxygen Purity

Regression Analysis

The regression equation is

Purity = 74.3 + 14.9 HC Level

Predictor	Coef	SE Coef	T	P
Constant	74.283 ← $\hat{\beta}_0$	1.593	46.62	0.000
HC Level	14.947 ← $\hat{\beta}_1$	1.317	11.35	0.000

S = 1.087 R-Sq = 87.7% R-Sq (adj) = 87.1%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	152.13	152.13	128.86	0.000
Residual Error	18	21.25 ← SS_E	1.18 ← $\hat{\sigma}^2$		
Total	19	173.38			

Predicted Values for New Observations

New Obs	Fit	SE Fit	95.0% CI	95.0% PI
1	89.231	0.354	(88.486, 89.975)	(86.830, 91.632)

Values of Predictors for New Observations

New Obs	HC Level
1	1.00

17

Example: Python Output

	coef	std err	t	P>|t|	[0.025	0.975]
	$b_0 = \hat{\beta}_0$					
Intercept	74.283	1.593	46.62	0.000	71.083	77.603
X	14.947	1.317	11.35	0.000	12.346	17.448
	$b_1 = \hat{\beta}_1$					

18

Distribution of b_0 ?

$$b_0 = \overline{y} - b_1\overline{x}$$

$$E(b_0) = \beta_0$$

$$V(b_0) = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)$$

$$b_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

19